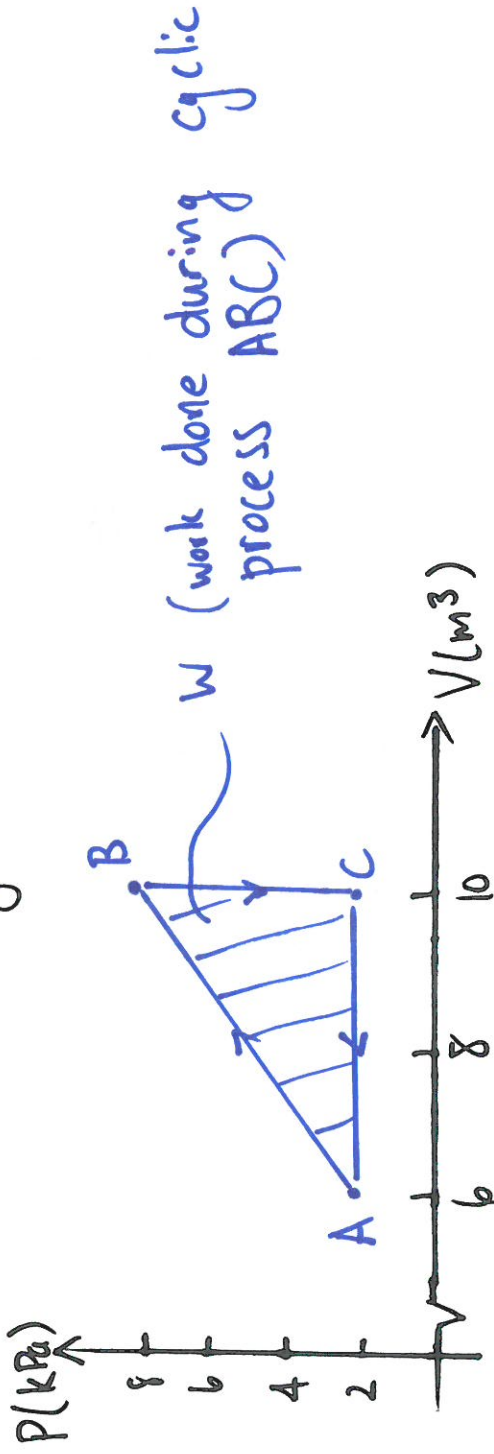


Problem 20.30 (Serway)



This is a cyclic process $\Rightarrow \Delta E_{int} = 0$ (we return to the initial state A)

(a) Since $\Delta E_{int} = 0$, from the first law it follows that

$$Q = -W$$

We can determine W from the shaded area:

$$\begin{aligned} W &= \frac{1}{2} (10 \text{ m}^3 - 6 \text{ m}^3) (8 \text{ kPa} - 2 \text{ kPa}) \\ &= \frac{1}{2} (4 \text{ m}^3) (6 \times 10^3 \text{ Pa}) \times \frac{1 \text{ N/m}^2}{1 \text{ Pa}} \\ &= 12 \times 10^3 \text{ N} \cdot \text{m} \times \frac{1 \text{ J}}{1 \text{ N} \cdot \text{m}} \\ &= 12 \text{ kJ} \end{aligned}$$

②
But the sign of W is negative, since for the cyclic process the area is larger for the expansion process ($A \rightarrow B$) than the compression process ($C \rightarrow A$).

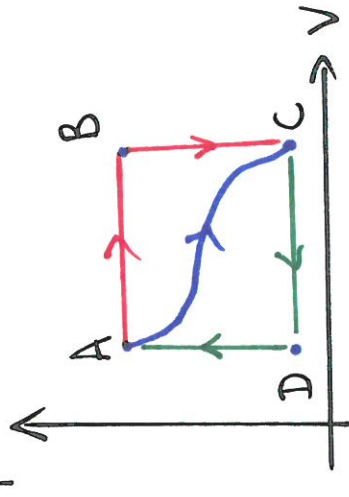
$$\therefore Q = -(-12 \text{ kJ})$$

$$Q = +12 \text{ kJ}$$

(b) If the cycle is reversed, W will be positive

$$\therefore Q = -W = -12 \text{ kJ}$$

Problem 20.40 (Serway)



$$\text{Given: } \Delta E_{\text{int}, AC} = +800 \text{ J}$$

$$W_{ABC} = -500 \text{ J}$$

(a) From conservation of energy, the change in internal energy is the same for paths $A \rightarrow C$ and $A \rightarrow B \rightarrow C$.
 $\Rightarrow \Delta E_{\text{int}, AC} = \Delta E_{\text{int}, ABC}$

From 1st law (for path ABC)

$$\Delta E_{\text{int}, ABC} = Q_{ABC} + W_{ABC}$$

$$\begin{aligned} Q_{ABC} &= \Delta E_{\text{int}, ABC} - W_{ABC} \\ &= +800 \text{ J} - (-500 \text{ J}) \end{aligned}$$

$$Q_{ABC} = +1300 \text{ J}$$

(b) Pressure at point A is five times that at point C ②

$$\Rightarrow P_C = \frac{1}{5} P_A \quad - ①$$

Work done from C $\rightarrow$ D is done at constant pressure

$$\therefore W_{CD} = -P_C \Delta V_{CD} \quad - ②$$

$$\text{but } \Delta V_{CD} = -\Delta V_{AB} \quad - ③$$

Using relations ① and ③ in eq. ②:

$$W_{CD} = \frac{1}{5} P_A \Delta V_{AB}$$

$$\begin{aligned} &= -\frac{1}{5} W_{AB} \\ &= -\frac{1}{5} (-500 \text{ J}) \\ W_{CD} &= +100 \text{ J} \end{aligned}$$

$\left\{ \begin{array}{l} W_{AB} = W_{BC} \text{ since } W_{BC} = 0 \\ W_{AB} = -P_A \Delta V_{AB} \end{array} \right. \downarrow \text{constant volume process.}$

③

(c) $W_{CDA} = W_{CD}$ Since $W_{DA} = 0$ (isovolumetric process)

1st law:
$$\begin{aligned} Q_{CDA} &= \Delta E_{int, CDA} - W_{CDA} \\ &= -\Delta E_{int, AC} - W_{CD} \\ &= -(800 \text{ J}) - (100 \text{ J}) \\ Q_{CDA} &= -900 \text{ J} \end{aligned}$$
 (heat released to the environment)

(d)
$$\begin{aligned} \Delta E_{int, DA} &= +500 \text{ J} \\ \Delta E_{int, CD} &= \Delta E_{int, CDA} - \Delta E_{int, DA} \\ &= -\Delta E_{int, AC} - \Delta E_{int, DA} \\ &= -(800 \text{ J}) - 500 \text{ J} \\ \Delta E_{int, CD} &= -1300 \text{ J} \end{aligned}$$

1st law:
$$\begin{aligned} Q_{CD} &= \Delta E_{int, CD} - W_{CD} \\ Q_{CD} &= -1300 \text{ J} - (100 \text{ J}) = -1400 \text{ J} \end{aligned}$$